

CHELTENHAM GIRLS HIGH SCHOOL



YEAR 11
COMMON TEST 4
2010

MATHEMATICS

Time allowed: 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are not of equal value.
- All **necessary working** should be shown in every question. Marks may be deducted for careless or badly arranged work
- Examination answers are to be completed in blue or black pen.
- Board approved calculators may be used.
- Each **question** is to be started on a **new page**.
- The marks allocated for each question are indicated

Name: _____

Class Teacher: _____

Question	1	2	3	4	Total	%
Mark	/14	/14	/15	/15	/58	

Question 1 (14 marks) Start a new page**Mark**

- (a) Write 8539.25377 correct to 3 significant figures 1
- (b) Factorise completely: $3x^2 - 12y^2$. 2
- (c) Express $\frac{\sqrt{32} - \sqrt{8}}{3\sqrt{2}}$ in the form of a simple fraction $\frac{a}{b}$, where a and b are rational numbers. 2
- (d) Solve the equation $\frac{2x}{x-5} = \frac{3}{7}$ 2
- (e) Sketch the graph of $y = |3 - 2x|$ and find the set of values of x for which $y \leq 3$. 3
- (f) Given that $3 \log_x 2 - \log_x 128 = 4$, find the value of x . 2
- (g) Find the **EXACT** value of:
- (i) $\sin 225^\circ$ 1
- (ii) $\sec 120^\circ$ 1

Question 2 (14 marks) Start a new page**Mark**

- (a) Differentiate:
- (i) $2x^3 + \frac{1}{2x^3}$ 2
- (ii) $\frac{(x+2)^2}{x}$ 2
- (iii) $(5x+1)^5$ 2
- (iv) $\sqrt{3x^2 - 1}$ 2
- (b) Calculate the gradient of the curve $y = \frac{3x^2 - 4}{3 - 2x}$ at the point (2, -8) 3
- (c) The curve $y = x^3 - 3x^2 - 9x + 7$ has two stationary points. Find the co-ordinates of the minimum turning point. 3

Question 3 (15 marks) Start a new page**Mark**

- (a) If α and β are the roots of $3x^2 + 2x - 3 = 0$, find the value of
- (i) $\alpha + \beta$ 1
 - (ii) $\alpha \beta$ 1
 - (iii) $(\alpha + 1)(\beta + 1)$ 2
- (b) Find the values of t for which the equation $x^2 + tx + 2t - 3 = 0$ has real roots. 3
- (c) Solve the equation $4^x - 9(2^x) + 8 = 0$ 3
- (d) Find all values of θ if $0 \leq \theta \leq 360^\circ$ for $\cos \theta = 0.5$ 2
- (e) Find, by completing the square method, the centre and the radius of the circle with equation $x^2 + y^2 = 8x - 4y + 29$. 3

Question 4 (15 marks) Start a new page**Mark**

- (a) Solve the equation $5^y = 10$, give your answer correct to 2 decimal places. 2
- (b) Given $2x^2 + 9x - 1 \equiv a(x+1)^2 + b(x+1) + c$, find the values of a , b and c . 3
- (c) A parabola P has equation $x^2 = -8(y - 4)$.
Draw a neat sketch of P . You must show clearly : 1
- (i) the co-ordinates of its vertex 1
 - (ii) the co-ordinates of its focus 1
 - (iii) the equation of its directrix 1
 - (iv) the x - intercepts 2
- (d) A and B are the points $(4, 2)$ and $(1, 5)$ respectively.
- (i) Find the equation of the locus of a point $Q(x, y)$ which moves such that QA is perpendicular to QB . 3
 - (ii) Describe the locus of Q in part (i) geometrically. 1

End of paper

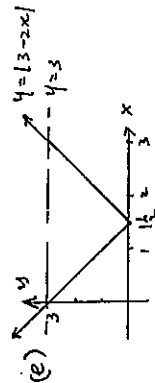
Q1. (a) 8540

(b) $3(x^2 - 4y^2)$
 $= 3(x + 2y)(x - 2y)$

(c) $\frac{4\sqrt{2} - 2\sqrt{2}}{3\sqrt{2}} = \frac{2}{3}$

$\therefore a = 2$ & $b = 3$

(d) $\frac{2x}{x-5} = \frac{3}{7}$
 $14x = 3x - 15$
 $x = -\frac{15}{11}$



$0 \leq x \leq 3$

(f) $\log_x \frac{2^3}{128} = 4$

$\log_x \frac{1}{16} = 4$

$x^4 = \frac{1}{16}$

$x = \frac{1}{2}$ only

(g) (i) $\sin 225^\circ = -\sin 45^\circ$
 $= -\frac{1}{\sqrt{2}}$

(ii) $\sec 120^\circ = \frac{1}{\cos 60^\circ}$
 $= -2$

Q2. (a) (i) $f' = 6x^2 - \frac{3}{2x^4}$

(ii) $f' = 1 - \frac{4}{x^2}$

(iii) $f' = 25(5x+1)^4$

(iv) $f' = \frac{3x}{\sqrt{3x^2-1}}$

(b) $y' = \frac{-6x^2 + 18x - 8}{(3-2x)^2}$

at $x=2$, $y' = \frac{-24+36-8}{1} = 4$

(c) $y' = 3x^2 - 6x - 9$
 $3x^2 - 6x - 9 = 0$
 $3(x-3)(x+1) = 0$
 $x=3$ or $x=-1$

at $x=3$

x	2	3	4
y'	-9	0	15

$\therefore$ Minimum at $(3, -20)$

(d) (i) $\alpha + \beta = -\frac{b}{a} = -\frac{2}{3}$

(ii) $\alpha\beta = \frac{c}{a} = -1$

(iii) $(\alpha+1)(\beta+1)$
 $= \alpha\beta + \alpha + \beta + 1 = -\frac{2}{3}$

(b) For real roots: $b^2 - 4ac \geq 0$
 $4^2 - 8c + 12 \geq 0$
 $(c-6)(c-2) \geq 0$
 $c \geq 6$ or $c \leq 2$

(c) let $u = 2^x$
 $u^2 - 9u + 8 = 0$
 $(u-8)(u-1) = 0$
 $u=8$ or $u=1$
 $2^x = 8$ or $2^x = 1$
 $x=3$ or $x=0$

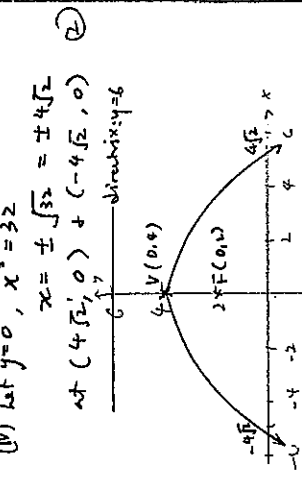
(d) $\cos \theta = 0.5$
 $\theta = 60^\circ$ or 300°

(e) $x^2 - 5x + 4^2 + y^2 + 4y + 2^2 = 25 + 4^2 + 2^2$
 $(x-4)^2 + (y-2)^2 = 49$
 $\therefore$ Centre at $(4, 2)$
 Radius = 7 units.

Q4. $5^y = 10$
 $\therefore y = \frac{\log_{10} 10}{\log_{10} 5} = 1.43 (2 \text{ d.p.})$

(b) $2x^2 + 9x - 1 = a(x^2 + (2a+b)x + a+bx)$
 $a=2$
 $b=5$
 $c=-8$

(c) $a=2$
 (i) $V(0, 4)$
 (ii) $F(0, 2)$
 (iii) direction: $y=6$
 (iv) let $y=0$, $x^2=32$
 $x = \pm \sqrt{32} = \pm 4\sqrt{2}$
 at $(4\sqrt{2}, 0)$ & $(-4\sqrt{2}, 0)$



(d) (i) $M_{PA} = \frac{y-2}{x-4}$, $M_{PB} = \frac{y-5}{x-1}$
 $M_{PA} \times M_{PB} = -1$
 $\frac{y-2}{x-4} \times \frac{y-5}{x-1} = -1$
 $(y-2)(y-5) = -(x-4)(x-1)$
 $x^2 + y^2 - 5x - 7y + 11 = 0$
 $(x-2\frac{1}{2})^2 + (y-3\frac{1}{2})^2 = 4\frac{1}{4}$

(ii) The locus of P is a circle
 Centre $(2\frac{1}{2}, 3\frac{1}{2})$
 Radius $\sqrt{4\frac{1}{4}}$ units.

